

Use Case “CUD Product”

1. Use case code

UC004

2. Brief Description

This use case describes the interaction between the **Product Manager** and the **AIMS system** when the actor wants to **create, update, or delete product information** in the system. The Product Manager can add new products, modify the information of an existing product, or delete products that are no longer available for sale. The system validates the input data and ensures that all business rules related to product management are satisfied before completing the operation.

3. Actors

3.1 Name of Actor

Product Manager

4. Preconditions

- The AIMS system is operating normally.
- The Product Manager has logged into the system with sufficient permissions to manage products.
- The Product Manager is currently on the **product management screen**.
- For update or delete operations, the selected product must already exist in the system.

5. Basic Flow of Events

1. The Product Manager accesses the product management screen.
2. The system displays the list of products currently stored in the system.
3. The Product Manager chooses one of the following operations: Create Product, Update Product, or Delete Product.
4. Depending on the selected operation:
 - **If Create Product is selected:**
 - 4.1a. The system displays a product creation form.
 - 4.2a. The Product Manager enters the required product information (including mandatory product type-specific details) and submits the request.
 - 4.3a. The system validates the input data according to the product rules.

- 4.4a. The system stores the product information in the database.
- **If Update Product is selected:**
 - 4.1b. The Product Manager selects an existing product to edit (only one product at a time), and the system retrieves the selected product information.
 - 4.2b. The system displays the product information and allows the Product Manager to modify it.
 - 4.3b. The Product Manager updates the product information and submits the changes.
 - 4.4b. The system validates the updated data and saves the changes to the database.
- **If Delete Product is selected:**
 - 4.1c. The Product Manager selects between 1 and up to 10 products to delete simultaneously and submits the request.
 - 4.2c. The system checks the Product Manager's daily deletion limit.
 - 4.3c. The system checks whether the stock quantity of the selected product(s) is zero.
 - 4.4c. Confirming the stock quantity equals zero, the system deletes the product from the system.
- 5. The system records the history of the product creation, update, or deletion operation.
- 6. The system displays a success notification confirming that the operation has been completed successfully.
- 7. The use case ends.

6. Alternative flows

No	Location	Condition	Action	Resume location
1	Step 4.3a, 4.4b	Input data does not comply with the required rules (e.g., missing mandatory fields, incorrect format)	The system notifies the Product Manager about the invalid input and asks for correction	Resume at Step 4.2a or 4.3b
2	Step 4.3a, 4.4b	The current price is not between 30% and 150% of the original value	The system displays an error message stating the price constraint violation and asks for adjustment	Resume at Step 4.2a or 4.3b

3	Step 4.1c	The Product Manager attempts to select more than 10 products to delete simultaneously	The system rejects the request and displays an error message	Resume at Step 4.1c
4	Step 4.2c	The Product Manager attempts to delete more than 20 products per day	The system rejects the request and notifies the manager that the daily deletion limit has been reached	Use case ends
5	Step 4.3c	The product stock quantity is greater than zero	The system changes the product status to "deactivated" instead of deleting the product	Resume at Step 5

7. Input data

Table A – Input data of CUD Product

No	Data fields	Description	Mandatory	Valid condition	Example
1	Barcode	Unique identifier of the product	Yes	Must not duplicate existing barcode	893523451234
2	Title	Product title	Yes	Non-empty text	One Piece Vol.1
3	Category	Product category	Yes	Must belong to supported types	Book
4	Original value	Original value of the product (without VAT)	Yes	Must be positive number	100000
5	Current price	Current selling price (without VAT)	Yes	Must be between 30% and 150% of original value	120000
6	Description	General product description	Yes	Text	New book
7	Dimensions	Height, width, length	Yes	Valid measurement	20x15x5 cm
8	Weight	Product weight	Yes	Positive number	1 kg
9	Quantity in stock	Number of items in stock	Yes	Integer ≥ 0	100
10	Product type specific information	Additional information depending on product type	Yes	Must follow product type rules	Author, Director
11	Reason for stock change	The explicit reason when stock needs to be adjusted manually	Yes	Non-empty text	"Added new items" or "Removed due to damage"

8. Output data

Table B – Output data of CUD Product

No	Data fields	Description	Display format	Example
1	Operation result	Notification of successful product creation, update, or deletion.	Text message	Product added successfully
2	Error / Validation message	Notification explaining why an operation is invalid (e.g., incorrect format, rule violation).	Text message	Invalid date format / Current price must be at least 30% of original value
3	Product ID	Identifier of the processed product.	Numeric / String	P001
4	Product title	Name of the product.	Text	One Piece Vol.1
5	Updated product information	Updated product data displayed after operation.	Structured text	Updated price: 120000
6	Product status	Availability status of the product.	Text	Available / Deactivated

9. Postconditions

- The product information is successfully created, updated, or deleted in the system, or its status is changed to "deactivated" if a deletion is attempted on a product with a stock quantity greater than zero.
- The system records the operation history of the product creation, update, or deletion.
- The product list is updated to reflect the latest changes.
- The Product Manager can continue managing other products.